Cultural Reasoning for Responsible AI-Assisted Behavior Change

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⅓ of our food is wasted - can AI help us waste less food?

"I have stale rye bread" 😩

Here are some suggested recipes...

These are great ideas! 😍

Would you like more suggestions?

2 hours later What was I even doing? 🤔

Current AI chatbots are

- ✅ Eloquent
- ✅ Knowledgeable
- ✅ Engaging

❌ But they do not motivate **action**

Current AI reasoning vs. real-world reasoning

Recent progress

Since 2024, “reasoning” has become a major frontier in LLMs

Models now solve complex math and logic problems via chain-of-thought

How reasoning is currently evaluated

Benchmarks: math, code, symbolic logic

Success = correct answer inside the conversation

What is missing

Reasoning about **unstated priors** (norms, practices, values)

Reasoning about **human adoption and interpretation**

+ Evaluating outcomes outside the conversation

Culture as Unstated Priors

Many assumptions that guide interpretation and behavior are rarely stated.

Examples:

- what foods are familiar
- what counts as effort or indulgence
- what metaphors imply socially
- what behaviors feel appropriate

These assumptions function as **priors**.

Culture can be modeled as the set of unstated priors that shape interpretation and behavior.

The problem with cultural NLP today

"Cultural NLP" is growing fast

Hershcovich et al. (2022)

Pawar et al. (2025)

Tends to **trivialize culture**:

Treats it as a fixed label ("Danish culture")

Reduces it to *stereotypes*

Can AI *interpret, reuse, and reason through culture?*

Literary interpretation surfaces unstated priors

If culture operates as unstated priors, **literary texts** are where these priors become visible. Why?

- Literature relies heavily on **shared background knowledge**
- Authors rarely explain the assumptions their readers already share
- Meaning emerges through **ambiguity, contrast, and interpretation**

Reasoning under different assumptions

Example: *Babettes Gæstebud* by Karen Blixen

A strict religious community receives an extravagant French dinner.

- community values modest living
- luxurious feast prepared for them
- villagers worry it may be sinful

Plausible interpretations:

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1. Strict-religion assumption
   pleasure distracts from faith → the feast is morally risky
2. Art-and-generosity assumption
   beauty and generosity reveal grace → the feast becomes transformative
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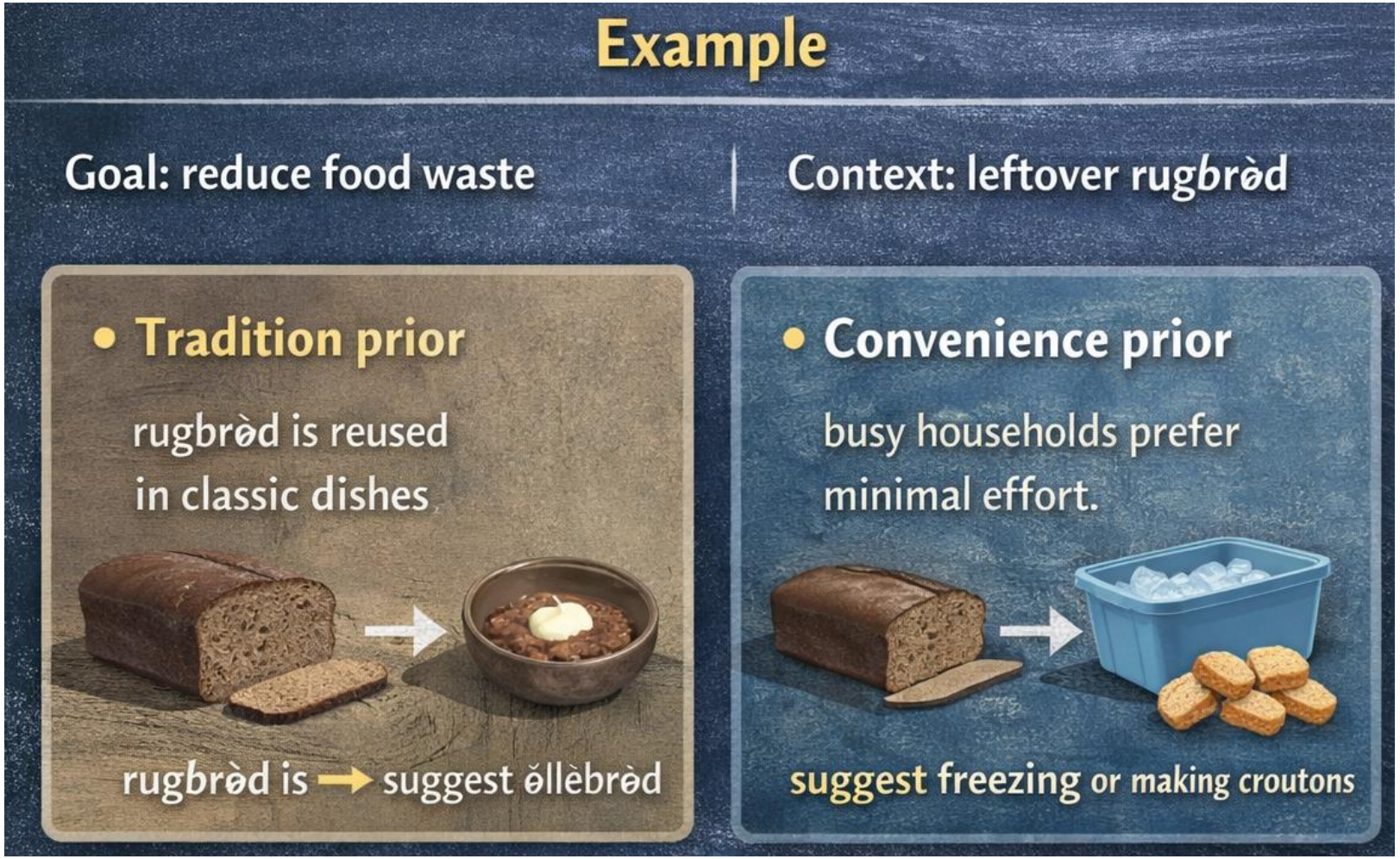
The story explores the tension between these assumptions.

Reasoning beyond the conversation

Two pieces of advice can both be **correct**.

But people may respond to them differently.



How can AI systems reason about unstated priors that shape how people interpret and act on advice?

Summary: Cultural Reasoning

Many real-world decisions depend on unstated priors — shared assumptions that shape how people interpret language and respond to advice.

Literary interpretation makes these priors visible, offering a rich setting to study how models reason about implicit cultural assumptions.

This raises a broader research question:
How can AI systems learn to reason about these priors — so that their outputs are not only correct, but meaningful and actionable in the real world?